MEDIUM

GEOMETRY AND TRIGONOMETRY

Right triangles and trigonometry

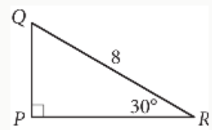
01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN GEOMETRY AND TRIGONOMETRY

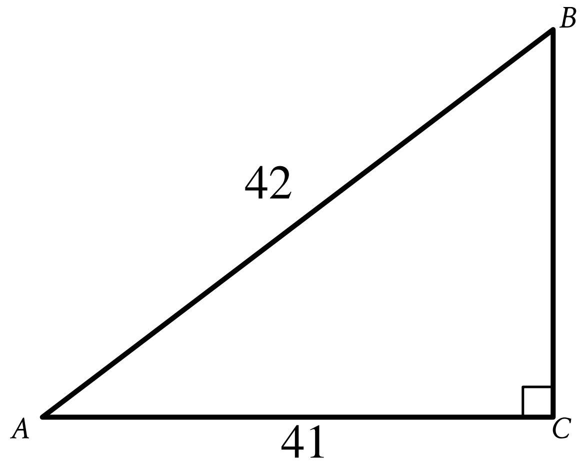


In the right triangle shown above, what is the length of $\overline{PQ}$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Angle upper A upper C upper B is a right angle.
- The length of side upper A upper B is 42.
- The length of side upper A upper C is 41.
- A note indicates the figure is not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

A. $\frac{42}{41}$

B. $\frac{41}{42}$

C. $\frac{1}{42}$

D. $\frac{1}{41}$

Q3

GEOMETRY AND TRIGONOMETRY

In right triangle RST, the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin R$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos S$?

A. $\frac{\sqrt{15}}{15}$

B. $\frac{\sqrt{15}}{4}$

C. $\frac{4\sqrt{15}}{15}$

D. $\sqrt{15}$

Q4

GEOMETRY AND TRIGONOMETRY

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\bar{BC}$ is 136 millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\bar{AB}$?

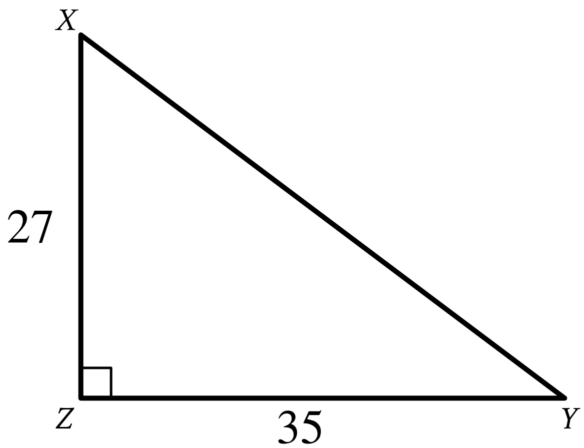
A. 34

B. 102

C. 136

D. 170

Q5



Note: Figure not drawn to scale.

- Angle upper Z is a right angle.
- The length of side upper X upper Z is 27.
- The length of side upper Y upper Z is 35.
- A note indicates the figure is not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

A. $\tan X$

B. $\tan Y$

C. $\cos X$

D. $\cos Y$

Q6 GEOMETRY AND TRIGONOMETRY

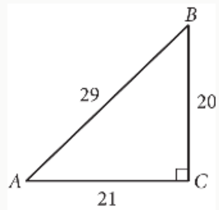
One leg of a right triangle has a length of 43.2 millimeters. The hypotenuse of the triangle has a length of 196.8 millimeters. What is the length of the other leg of the triangle, in millimeters?

A. 43.2

B. 120

C. 192

D. 201.5



In the figure above, what is the value of $\tan(A)$?

A. $\frac{20}{29}$

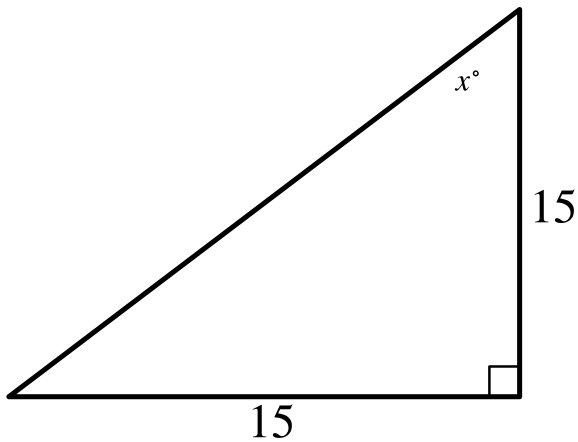
B. $\frac{21}{29}$

C. $\frac{20}{21}$

D. $\frac{21}{20}$

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x°.
- The lengths of the 2 sides adjacent to the right angle are as follows:
  - 15
  - 15
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of x?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

9 9 9 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GEOMETRY AND TRIGONOMETRY

A right triangle has legs with lengths of 28 centimeters and 20 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

A. $8\sqrt{6}$

B. $4\sqrt{74}$

C. 48

D. 1,184

Q10

GEOMETRY AND TRIGONOMETRY

The length of a rectangle’s diagonal is $5\sqrt{17}$, and the length of the rectangle’s shorter side is 5. What is the length of the rectangle’s longer side?

A. $\sqrt{17}$

B. 20

C. $15\sqrt{2}$

D. 400